

Flexible Formation Fencing Control of Second-Order Multi-Agent Systems

Liwei Kou^{1,2}, Long Jian², Ying Xue³, Guangyu Zuo^{1,2}, Yi Huang^{4,5}, Yinke Dou^{1,2}

1. Shanxi Energy Internet Research Institute, Taiyuan 030032, P. R. China

E-mail: {kouliwei, zuoguangyu, douyinke}@tyut.edu.cn

2. College of Electrical and Power Engineering, Taiyuan University of Technology, Taiyuan 030024, P. R. China

E-mail: jianlong@tyut.edu.cn

3. State Grid Shanxi Electric Power Marketing Service Center, Taiyuan 030008, P. R. China

E-mail: 771391876@qq.com

4. Institute of Artificial Intelligence and Advanced Computing, Hunan University of Technology and Business, Changsha 410205, P. R. China

5. Shandong New Generation Information Industry Technology Research Institute Ltd, Inspur Group Company Ltd, Jinan 250100, P. R. China

E-mail: yihcsu@outlook.com

Abstract: This paper proposes a fencing control strategy with a flexible formation to address the fencing control problem of second-order multi-agent systems for a moving target. Specifically, regarding the target moving with an unknown time-varying velocity, a sliding mode estimator accounts to estimate the targets velocity in finite time. In addition, a barrier function is used to avoid collision between agents. Neither the global position measurement nor the fixed fencing formation is required for controller design. It is proven that the controller ensures that the target can be fenced into the convex hull of multi-agent systems, and collision between agents is avoided. Finally, the simulation results verify the efficacy of the proposed controller.

Key Words: Cooperative Fencing, Multi-agent System, Collision Avoidance, Velocity Estimate

1 Introduction

Over the past few decades, cooperative control of multi-agent system have been an active area of research. Compared with a single agent, multi-agent systems can perform complex tasks by information exchange and behavior cooperation among agents[1–5]. Nowadays, cooperative control of multi-agent system have been widely applied in many fields, such as unmanned air vehicles (UAVs), unmanned surface vehicles(USVs), autonomous underwater vehicles(AUVs), and so on[6–9].

Technically speaking, cooperative control of multi-agent systems mainly includes consensus and synchronization [10, 11], formation control [12], containment control [13, 14], to name a few. In this paper, we investigate the fencing control of multi-agent systems, where a cooperative controller is designed to fence a target of interest into the convex hull formed by multi-agent systems [15]. The fencing control of multi-agent systems can be used to monitor the target's surrounding environment for guaranteeing the safety operation, which is of more practical potentials in military and civil applications.

Nowadays, there are some research results regarding the fencing control of multi-agent systems. According to the different formation shapes, the fencing control problem can be divided into circular fencing, elliptical fencing, safe convex-fencing, and convex fencing, which are illustrated in Fig. 1. The circular fencing aims to design the controller to drive the agents to orbit around a target along a common circle. In [16], the authors studied the circular fencing problem of non-

holonomic multi-agent systems under a directed ring graph. Later, the results were further extended to address the circular fencing problem for multiple AUVs in [17]. In [18], the authors investigated the circular fencing control for multiple mobile robots in the presence of external disturbances. In [19], control schemes using only bearing measurements were proposed for the underactuated USVs. In practice, multi-agent systems are required to fence different areas for some specific applications. The circular fencing strategy will increase the collision risk among agents, and is not applicable from a safety perspective [20]. Much efforts have been made to solve the fencing control problem with others formations. For example, the authors in [21] proposed the elliptical fencing controller for single-integrator and double-integrator multi-agent systems by similarity transformation. In [22], the elliptical fencing controller was developed for multiple static targets and moving targets. Considering the more complex fencing area, the authors in [23] proposed a safe convex-fencing controller to drive multi-agent systems to move along the area with a minimum convex form. Such the controller makes the fencing formation more compact and can reduce energy consumption.

However, in the aforementioned references, multi-agent systems move around the target during performing the fencing task. In some specific applications, such as multiple USVs convoying the flying target, multi-agent systems only need to fence the target but not move around it, as shown in Fig. 1(d). In [24], multi-agent systems fenced the target with a cube formation in three-dimension space. The authors in [25] proposed a fencing controller with a regular polygon formation.

Most of the aforementioned literatures assume that the fencing formation is predefined before controller design.

This work was supported by Fundamental Research Program of Shanxi Province (202103021223048, 20210302124552) and the Natural Science Foundation for Young Scientists of Shanxi Agricultural University (2020QC15)

However, in the presence of dynamically changing environments or real-time fencing requirements, it is of great challenge to predefine a specific fencing formation. Based on this consideration, the authors in [15] proposed a novel flexible formation fencing controller for single-integrator multi-agent systems fencing the static target, where no any fencing formation needs to be predefined. The result was further extended to address the cases where single-integrator multi-agent systems fence the moving target in [26] and double-integrator multi-agent systems fence the moving target in [27] with a velocity estimator. The authors in [28] developed a two-level distributed fencing controller, which was tested on a real multi-USV experimental platform.

Motivated by the previous research on flexible fencing, we propose a fencing control strategy with a flexible formation to address the fencing control problem of second-order multi-agent systems for a moving target. The aforementioned works on flexible fencing assume that the target's velocity is constant. In the set-up adopted in the paper, the target of interest moves with an unknown time-varying velocity, which is more applicable to the practical scenarios. Particularly, a finite-time sliding mode (FTSM) estimator is introduced to estimate the target's velocity. Then, a fencing controller is developed such that the target is fenced into the convex hull formed by second-order multi-agent systems. In addition, a barrier function accounts to avoid collision among agents.

Notation: Let $\mathbb{N} = \{1, 2, \dots, N\}$ be the set of multi-agent systems, where $N \geq 3$ is the agents number; A n -dimension real vector is denoted by $\mathbf{x} \in \mathbb{R}^n$, whose distance norm is calculated by $\|\mathbf{x}\| = (\sum_{i=1}^n x_i^2)^{1/2}$ which x_i being the i -th element of $\mathbf{x}$; $\text{sig}^\alpha(\mathbf{x}) = [|x_1|^\alpha \text{sign}(x_1), \dots, |x_n|^\alpha \text{sign}(x_n)]^T$, $\text{sign}(\mathbf{x}) = [\text{sign}(x_1), \dots, \text{sign}(x_n)]^T$; For any two vectors $\mathbf{x}_i$ and $\mathbf{x}_j$, $\mathbf{x}_{ij} := \mathbf{x}_i - \mathbf{x}_j$; $\mathbf{1}_N$ is an $N \times 1$ vector whose elements are all 1 and $\otimes$ denotes the Kronecker product.

2 Problem Formulation

Consider the following second-order multi-agent systems

$$\begin{aligned} \dot{\mathbf{x}}_i(t) &= \mathbf{v}_i(t) \\ \dot{\mathbf{v}}_i(t) &= \mathbf{u}_i(t), \quad i \in \mathbb{N} \end{aligned} \quad (1)$$

where $\mathbf{x}_i(t) \in \mathbb{R}^2$, $\mathbf{v}_i(t) \in \mathbb{R}^2$, and $\mathbf{u}_i(t) \in \mathbb{R}^2$ denote the position, velocity, and control input of agent i , respectively.

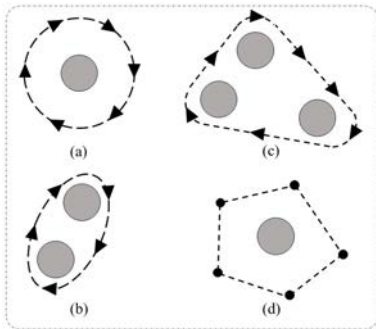


Fig. 1: Typical fencing modes. (a) Circular fencing; (b) Elliptical fencing; (c) Safe convex-fencing; (d) Convex fencing.

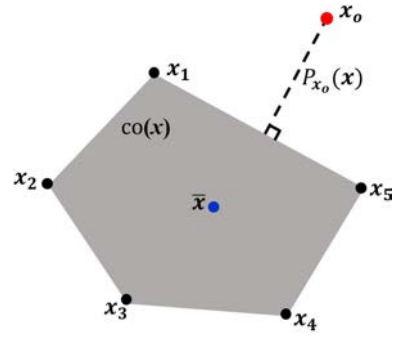


Fig. 2: Illustration of the convex hull $\text{co}(\mathbf{x})$ (the gray polygon), the center of the convex hull $\bar{\mathbf{x}}$ (the blue dot), the target $\mathbf{x}_o$ (the red dot), and $P_{\mathbf{x}_o}(\mathbf{x})$ (the dash line).

The notation (t) is omitted for simplicity. Define the position distribution as $\mathbf{x} = [\mathbf{x}_1^T, \mathbf{x}_2^T, \dots, \mathbf{x}_N^T]^T$. The related convex is described as

$$\text{co}(\mathbf{x}) = \left\{ \sum_{i \in \mathbb{N}} \rho_i \mathbf{x}_i : \rho_i \geq 0 \text{ and } \sum_{i \in \mathbb{N}} \rho_i = 1 \right\}$$

The center of the convex hull $\text{co}(\mathbf{x})$ is $\bar{\mathbf{x}} = \frac{1}{N} \sum_{i \in \mathbb{N}} \mathbf{x}_i$. It is easily observed that $\bar{\mathbf{x}} \in \text{int}(\text{co}(\mathbf{x}))$.

The target dynamics is denoted by

$$\begin{aligned} \dot{\mathbf{x}}_o &= \mathbf{v}_o \\ \dot{\mathbf{v}}_o &= \mathbf{u}_o \end{aligned} \quad (2)$$

where $\mathbf{x}_o$ denotes the target's position, $\mathbf{v}_o$ and $\mathbf{u}_o$ are the time-varying unknown velocity and acceleration, respectively.

Before formulating the fencing problem, we introduce the distance between $\mathbf{x}_o$ and the convex hull $\text{co}(\mathbf{x})$ as

$$P_{\mathbf{x}_o}(\mathbf{x}) = \inf_{\boldsymbol{\eta} \in \text{co}(\mathbf{x})} \|\boldsymbol{\eta} - \mathbf{x}_o\| \quad (3)$$

It is observed that $P_{\mathbf{x}_o}(\mathbf{x}) = 0$ iff $\mathbf{x}_o \in \text{co}(\mathbf{x})$, which indicates that multi-agent systems (1) achieve the target fencing task. For ease to understand, Fig. 2 illustrates the $\text{co}(\mathbf{x})$, $\mathbf{x}_o$, $\bar{\mathbf{x}}$, and $P_{\mathbf{x}_o}(\mathbf{x})$ with a simple example.

In the paper, the neighbors of agent i are defined using the nearest neighbor rule. Particularly, the neighbors are defined as

$$\mathcal{N}_i = \{j \in \mathbb{N}, j \neq i : \|\mathbf{x}_{ij}\| \leq \mu\} \quad (4)$$

where μ is the sensing distance and satisfies $\mu > d > 0$ with d being the collision distance.

To solve the target fencing problem, we need the following assumptions.

Assumption 1. At the initial time t_0 , there exists no collision avoidance, i.e., $\|\mathbf{x}_{ij}(t_0)\| > d, \forall i \neq j \in \mathbb{N}$.

Assumption 2. The acceleration of the target (2) satisfies Lipschitz continuity.

Remark 1. This assumption is necessary for the design of the sliding mode estimator, which will be introduced later. On the other hand, this assumption is reasonable due to the fact that the input of a control system often suffers from the saturation and jerk constraints from a practical perspective.

Then, the flexible formation target fencing problem is formulated as follows.

Flexible formation target fencing: Consider the second-order multi-agent system (1) and the target (2), design the fencing controller $\mathbf{u}_i$ with the finite-time sliding mode estimator $\hat{\mathbf{v}}_i$ such that the whole system satisfies the following properties.

- (P1) The target is fenced into the convex hull formed by the multi-agent system, i.e., $\lim_{t \rightarrow \infty} P_{\mathbf{x}_o(t)} \mathbf{x}(t) = 0$;
- (P2) Both the agents' velocity and the velocity estimator converge to the target's velocity $\mathbf{v}_o$, i.e., $\lim_{t \rightarrow \infty} \hat{\mathbf{v}}_i(t) = \mathbf{v}_o$, $\lim_{t \rightarrow \infty} \mathbf{v}_i(t) = \mathbf{v}_o$, $\forall i \in \mathbb{N}$;
- (P3) The collision between any pair of agents is avoided, i.e., $\|\mathbf{x}_{ij}(t)\| > d$, $\forall t \geq t_0$, $i \neq j \in \mathbb{N}$.

3 Main Results

Since the target's time-varying velocity is unavailable to all the agents, a finite time sliding mode estimator is firstly introduced to estimate the target's velocity. Then, we develop a control strategy for the flexible formation target fencing problem and provide rigorous analysis for convergence.

Define an error vector as $\boldsymbol{\sigma}_i = \mathbf{x}_o - \mathbf{x}_i$ with the time derivative being $\dot{\boldsymbol{\sigma}}_i = \mathbf{v}_o - \mathbf{v}_i$. The finite time sliding mode estimator is proposed as follows.

$$\begin{aligned} \dot{\mathbf{z}}_i &= \boldsymbol{\alpha}_{0,i} - \mathbf{v}_i \\ \boldsymbol{\alpha}_{0,i} &= -\lambda_0 L^{1/3} \text{sig}^{2/3}(\mathbf{z}_i - \boldsymbol{\sigma}_i) + \hat{\mathbf{v}}_i \\ \dot{\hat{\mathbf{v}}}_i &= \boldsymbol{\alpha}_{1,i} \\ \boldsymbol{\alpha}_{1,i} &= -\lambda_1 L^{1/2} \text{sig}^{1/2}(\hat{\mathbf{v}}_i - \boldsymbol{\alpha}_{0,i}) + \hat{\mathbf{u}}_i \\ \dot{\hat{\mathbf{u}}}_i &= -\lambda_2 L \text{sign}(\hat{\mathbf{u}}_i - \boldsymbol{\alpha}_{1,i}) \end{aligned} \quad (5)$$

where $\mathbf{z}_i$, $\boldsymbol{\alpha}_{0,i}$, $\boldsymbol{\alpha}_{1,i}$ are the auxiliary vectors, L is the Lipschitz constant of $\mathbf{u}_o$ in (2), and $\hat{\mathbf{v}}_i$ and $\hat{\mathbf{u}}_i$ are the estimate of $\mathbf{v}_o$ and $\mathbf{u}_o$, respectively. As proved in [29], when $\lambda_0 = 2$, $\lambda_1 = 1.5$, $\lambda_2 = 1$, the estimator (5) ensures that $\hat{\mathbf{v}}_i$ converges to $\mathbf{v}_o$ and $\hat{\mathbf{u}}_i$ converges to $\mathbf{u}_o$ in finite time.

Based on the estimator (5), the flexible formation fencing controller is proposed as follow.

$$\begin{aligned} \mathbf{u}_i &= \boldsymbol{\varphi}_i + k_2(\mathbf{x}_o - \mathbf{x}_i) + k_1(\hat{\mathbf{v}}_i - \mathbf{v}_i) + \hat{\mathbf{u}}_i \\ \boldsymbol{\varphi}_i &= \sum_{j \in \mathcal{N}_i} \alpha(\|\mathbf{x}_{ij}\|) \frac{\mathbf{x}_{ij}}{\|\mathbf{x}_{ij}\|} \end{aligned} \quad (6)$$

where $k_1 > 0$, $k_2 > 0$. The term $\boldsymbol{\varphi}_i$ represents the repulsive force between agent i and its neighbors $\mathcal{N}_i$. $\alpha(\|\mathbf{x}_{ij}\|)$ is a barrier function with the following form

$$\alpha(\|\mathbf{x}_{ij}\|) = \frac{1}{\|\mathbf{x}_{ij}\| - d} - \frac{1}{\mu - d} \quad (7)$$

When $\|\mathbf{x}_{ij}\| \rightarrow d^+$, we have $\alpha(\|\mathbf{x}_{ij}\|) \rightarrow +\infty$. It implies that if two agents are close enough, the repulsive force between them will tend to infinity for collision avoidance.

Then, we give the formal theorem for flexible formation fencing control as follows.

Theorem 1. Consider the multi-agent system (1) and the target (2) with controller (6) and estimator (5). Under Assumptions 1 and 2, the formation fencing control problem is solved in the sense of (P1), (P2), and (P3).

Proof: Define the position error as $\tilde{\mathbf{x}}_i = \mathbf{x}_i - \mathbf{x}_o$, the average position error as $\hat{\mathbf{x}} = \sum_{i=1}^N \tilde{\mathbf{x}}_i / N$. The second derivative of $\hat{\mathbf{x}}$ with respect to time is

$$\ddot{\hat{\mathbf{x}}} = \frac{1}{N} \sum_{i \in \mathbb{N}} (\mathbf{u}_i - \mathbf{u}_o)$$

Substituting (6) into the above equation yields

$$\begin{aligned} \ddot{\hat{\mathbf{x}}} &= \frac{1}{N} \sum_{i \in \mathbb{N}} \boldsymbol{\varphi}_i - k_2 \hat{\mathbf{x}} + \frac{1}{N} \sum_{i \in \mathbb{N}} (k_1(\hat{\mathbf{v}}_i - \mathbf{v}_i) + (\hat{\mathbf{u}}_i - \mathbf{u}_o)) \\ &= \frac{1}{N} \sum_{i \in \mathbb{N}} \boldsymbol{\varphi}_i - k_2 \hat{\mathbf{x}} - k_1 \dot{\hat{\mathbf{x}}} + \frac{1}{N} \sum_{i \in \mathbb{N}} (k_1(\hat{\mathbf{v}}_i - \mathbf{v}_o) + (\hat{\mathbf{u}}_i - \mathbf{u}_o)) \end{aligned}$$

As $j \in \mathcal{N}_i \Leftrightarrow i \in \mathcal{N}_j$ and $\mathbf{x}_{ij} = -\mathbf{x}_{ji}$, we have that $\sum_{i \in \mathbb{N}} \boldsymbol{\varphi}_i = \mathbf{0}$. Then

$$\ddot{\hat{\mathbf{x}}} + k_1 \dot{\hat{\mathbf{x}}} + k_2 \hat{\mathbf{x}} = \frac{1}{N} \sum_{i \in \mathbb{N}} (k_1(\hat{\mathbf{v}}_i - \mathbf{v}_o) + (\hat{\mathbf{u}}_i - \mathbf{u}_o))$$

This equation is input-state stable (ISS) as $k_1 > 0$ and $k_2 > 0$. According to Eq. (5), we have that $\hat{\mathbf{v}}_i - \mathbf{v}_o$ and $\hat{\mathbf{u}}_i - \mathbf{u}_o$ converge to zero in finite time. It implies that $\hat{\mathbf{x}}$ converge to zero as $t \rightarrow \infty$, i.e., $\lim_{t \rightarrow \infty} \hat{\mathbf{x}}(t) = \mathbf{0}$. It, together with the facts that $\bar{\mathbf{x}} = \frac{1}{N} \sum_{i \in \mathbb{N}} \mathbf{x}_i = \hat{\mathbf{x}} + \mathbf{x}_o$, indicates that $\lim_{t \rightarrow \infty} P_{\mathbf{x}_o} \mathbf{x}(t) = 0$. Thus, (P1) is proved.

Before proving (P2), we define some vectors as follows.

$$\begin{aligned} \mathbf{v} &= [\mathbf{v}_1^T, \dots, \mathbf{v}_N^T]^T \\ \hat{\mathbf{v}} &= [\hat{\mathbf{v}}_1^T, \dots, \hat{\mathbf{v}}_N^T]^T \\ \mathbf{u} &= [\mathbf{u}_1^T, \dots, \mathbf{u}_N^T]^T \\ \boldsymbol{\varphi} &= [\boldsymbol{\varphi}_1^T, \dots, \boldsymbol{\varphi}_N^T]^T \\ \hat{\mathbf{u}} &= [\hat{\mathbf{u}}_1^T, \dots, \hat{\mathbf{u}}_N^T]^T. \end{aligned}$$

A Lyapunov function candidate is constructed as

$$\begin{aligned} V &= \frac{1}{2} \sum_{i \in \mathbb{N}} \sum_{j \in \mathcal{N}_i} \int_{\|\mathbf{x}_{ij}\|}^{\mu} \alpha(s) ds + \frac{k_2}{2} \|\mathbf{x} - \mathbf{x}_o \otimes \mathbf{1}_N\|^2 \\ &\quad + \frac{1}{2} \|\mathbf{v} - \mathbf{v}_o \otimes \mathbf{1}_N\|^2 \end{aligned} \quad (8)$$

The partial differentiations of V with respect to $\mathbf{x}$, $\mathbf{x}_o \otimes \mathbf{1}_N$, $\mathbf{v}$, and $\mathbf{v}_o \otimes \mathbf{1}_N$ are, respectively

$$\begin{aligned} \frac{\partial V}{\partial \mathbf{x}^T} &= -(\boldsymbol{\varphi} + k_2(\mathbf{x}_o \otimes \mathbf{1}_N - \mathbf{x}))^T \\ \frac{\partial V}{\partial (\mathbf{x}_o \otimes \mathbf{1}_N)^T} &= k_2(\mathbf{x}_o \otimes \mathbf{1}_N - \mathbf{x})^T \\ \frac{\partial V}{\partial \mathbf{v}^T} &= (\mathbf{v} - \mathbf{v}_o \otimes \mathbf{1}_N)^T \\ \frac{\partial V}{\partial (\mathbf{v}_o \otimes \mathbf{1}_N)^T} &= -(\mathbf{v} - \mathbf{v}_o \otimes \mathbf{1}_N)^T \end{aligned}$$

Then, the time derivative of V is

$$\begin{aligned}\dot{V} &= \frac{\partial V}{\partial \mathbf{x}^T} \dot{\mathbf{x}} + \frac{\partial V}{\partial (\mathbf{x}_o \otimes \mathbf{1}_N)^T} (\dot{\mathbf{x}}_o \otimes \mathbf{1}_N) + \frac{\partial V}{\partial \mathbf{v}^T} \dot{\mathbf{v}} \\ &\quad + \frac{\partial V}{\partial (\mathbf{v}_o \otimes \mathbf{1}_N)^T} (\dot{\mathbf{v}}_o \otimes \mathbf{1}_N) \\ &= -(\boldsymbol{\varphi} + k_2(\mathbf{x}_o \otimes \mathbf{1}_N - \mathbf{x}))^T \mathbf{v} + k_2(\mathbf{x}_o \otimes \mathbf{1}_N - \mathbf{x})^T \\ &\quad \times (\mathbf{v}_o \otimes \mathbf{1}_N) + (\mathbf{v} - \mathbf{v}_o \otimes \mathbf{1}_N)^T \mathbf{u} - (\mathbf{v} - \mathbf{v}_o \otimes \mathbf{1}_N)^T \\ &\quad \times (\mathbf{u}_o \otimes \mathbf{1}_N) \\ &= -(\boldsymbol{\varphi} + k_2(\mathbf{x}_o \otimes \mathbf{1}_N - \mathbf{x}))^T (\mathbf{v} - \mathbf{v}_o \otimes \mathbf{1}_N) \\ &\quad + (\mathbf{v} - \mathbf{v}_o \otimes \mathbf{1}_N)^T (\mathbf{u} - \mathbf{u}_o \otimes \mathbf{1}_N) \\ &= -k_1 \|\mathbf{v} - \mathbf{v}_o \otimes \mathbf{1}_N\|^2 + h(t)\end{aligned}$$

where

$$h(t) = (\mathbf{v} - \mathbf{v}_o \otimes \mathbf{1}_N)^T (k_1(\hat{\mathbf{v}} - \mathbf{v}_o \otimes \mathbf{1}_N) + \hat{\mathbf{u}} - \mathbf{u}_o \otimes \mathbf{1}_N).$$

Let $W = V - \int_0^t h(\tau) d\tau$. Since both $\hat{\mathbf{v}}_i - \mathbf{v}_o$ and $\hat{\mathbf{u}}_i - \mathbf{u}_o$ converge to zero in finite time, $\int_0^t h(\tau) d\tau$ exists and hence W is lower bounded. From the above equation we have $\dot{W} = -k_1 \|\mathbf{v} - \mathbf{v}_o \otimes \mathbf{1}_N\|^2 \leq 0$. According to the invariant set theorem, the velocity of multi-agent systems converges to the target's time-varying velocity, i.e., $\lim_{t \rightarrow \infty} \mathbf{v}_i(t) = \mathbf{v}_o$. Thus, (P2) is proved.

Finally, contradiction is used to prove (P3). If the collision between agents i and j occurs, then $\exists T_0 > 0$ such that $\|\mathbf{x}_i(T_0) - \mathbf{x}_j(T_0)\| = d$ at the first time for $i \neq j \in \mathbb{N}$. As such

$$\begin{aligned}\int_{\|\mathbf{x}_{ij}(t)\|}^{\mu} \alpha(s) ds &= \int_d^{\mu} \left(\frac{1}{s-d} - \frac{1}{\mu-d} \right) ds \\ &= \ln(s-d) \Big|_d^{\mu} - 1 = \infty\end{aligned}$$

which implies a contradiction with the boundedness of W . Thus, the collision between any pair of agents is avoided and (P3) is proved. This completes the proof.

Remark 2. In [15, 26–28], it was assumed that the target is static or moves with an unknown constant velocity, which rapidly limits the application. While we consider the case where the target moves with an unknown time-varying velocity by introducing a finite time sliding mode estimator (5).

4 Simulations

In this part, a simulation example is presented to illustrate Theorem 1. This example includes five agents and a target. The initial position of five agents is set as

$$\mathbf{x}_i(t_0) = 20 \times [\cos(2(i-1)\pi/5), \sin(2(i-1)\pi/5)]^T \text{m}$$

The initial velocity $\mathbf{v}_i(t_0) = \mathbf{0}$ m/s, $i \in \mathbb{N}$. The initial target's position $\mathbf{x}_o(t_0) = [10, 30]^T$ m and the moving trajectory of the target is described as

$$\mathbf{x}_o(t) = [10 + 3t, 30 + 30 \sin(\pi/25t)]^T \text{m} \quad (9)$$

which means that $\mathbf{v}_o(t) = [3, \frac{6\pi}{5} \cos(\pi/25t)]^T$ m/s and $\mathbf{u}_o(t) = [0, -\frac{6\pi^2}{125} \sin(\pi/25t)]^T$ m/s². The controller parameters are specified as $k_1 = 5, k_2 = 6.75, k_3 = 0.5$. The parameters of $\alpha(s)$ in Eq. (7) are $\mu = 14$ and $d = 10$.

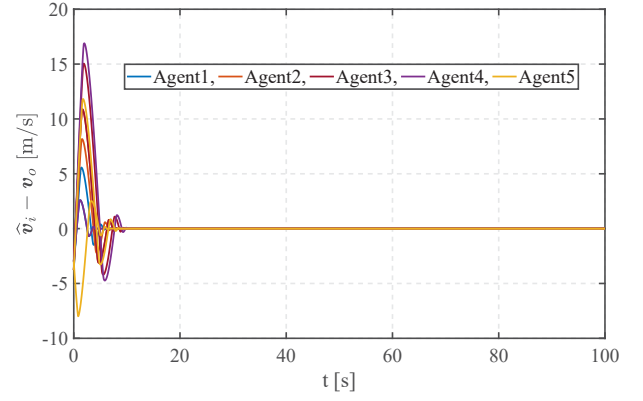


Fig. 3: Time evolution of the velocity estimate error under estimator (5).

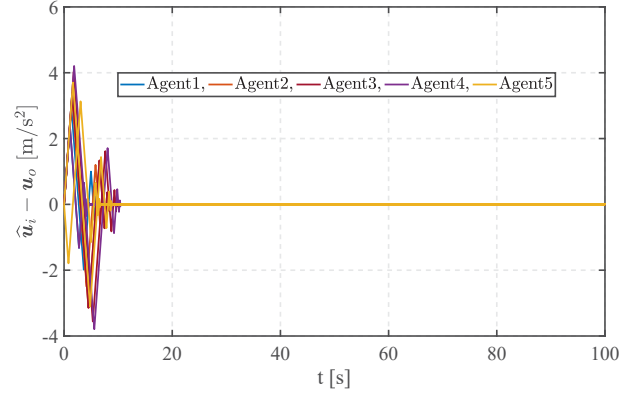


Fig. 4: Time evolution of the acceleration estimate error under estimator (5).

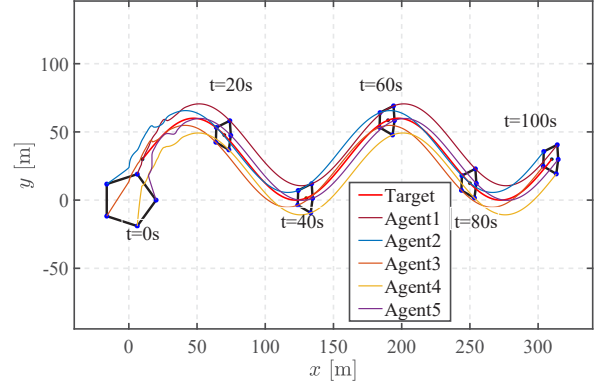


Fig. 5: Trajectories of multi-agents during target fencing control under the controller (6).

Fig. 3~Fig. 8 show the simulation results of the flexible formation fencing control under the controller (6). In particular, Fig. 3 and Fig. 4 show, respectively, the time evolution of the velocity estimate error $\hat{\mathbf{v}}_i - \mathbf{v}_o$ and acceleration estimate error $\hat{\mathbf{u}}_i - \mathbf{u}_o$. Note that both $\hat{\mathbf{v}}_i - \mathbf{v}_o$ and $\hat{\mathbf{u}}_i - \mathbf{u}_o$ converge to zero in finite time, which means that the target's unknown time-varying velocity and acceleration can be obtained by multi-agent systems in finite time. Fig. 5 depicts the trajectories of multi-agent systems during target fencing control under the controller (6). Note that although the initial formation of multi-agents is designed as a regular pentagon, it is transformed into another pentagon since no any formation is predefined before controller design. As shown in Fig.

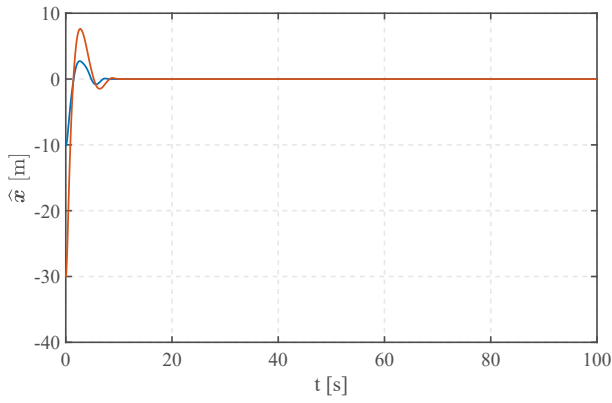


Fig. 6: Time evolution of the relative distance under the controller (6).

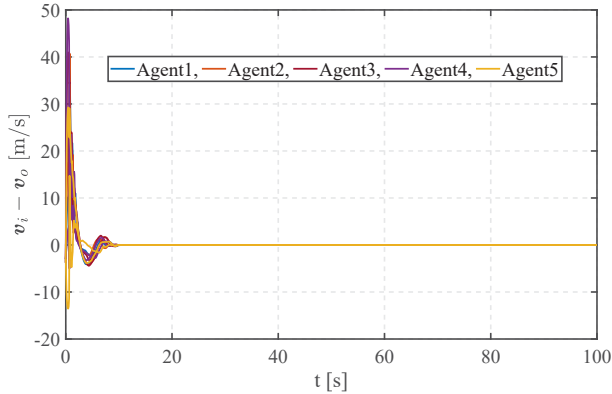


Fig. 7: Time evolution of the velocity error under the controller (6).

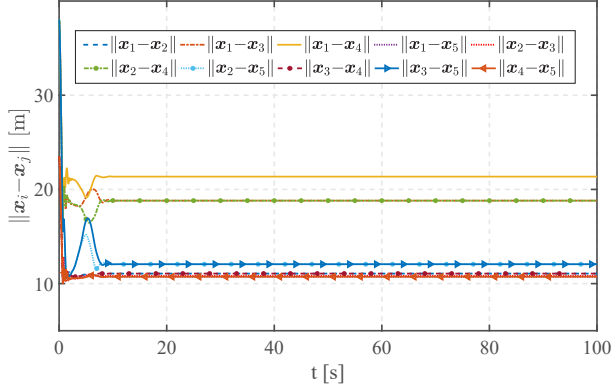


Fig. 8: Time evolution of the relative distance between any two agents under the controller (6).

6, the average position error exponentially converges to zero, which indicates that the target is fenced into the convex hull of multi-agents as described by (P1). Fig. 7 plots the time evolution of the velocity error. It is easily observed that the multi-agent systems' velocity v_i converges to the target's velocity v_o as expected by (P2). Finally, the relative distance between any two agents, i.e., $\|x_i - x_j\|$, is depicted in Fig. 8. It is shown that $\|x_i - x_j\| > d$ is always satisfied, which validates the collision avoidance described by the property (P3).

5 Conclusions

In this paper, we have proposed a control strategy to address the flexible formation fencing control problem of

multi-agent systems, such that the target with an unknown time-varying velocity is fenced into the convex hull formed by multi-agent systems. A sliding mode estimator has been introduced to simultaneously estimate the target's unknown time-varying velocity and acceleration in finite time. In addition, collision avoidance is avoided during target fencing. It is challenging to extend the results to more complicated nonlinear systems such as AUVs [30, 31], which will be further investigated in the future.

References

- [1] K. Oh, M. Park, and H. Ahn, A survey of multiagent formation control. *Automatica*, 53: 424–440, 2015.
- [2] Y. Hong and C. Zhai, Dynamic coordination and distributed control design of multi-agent systems. *Control Theory & Applications*, 28(10): 1506–1512, 2011.
- [3] J. Qin, Q. Ma, Y. Shi, et al, Recent advances in consensus of multi-agent systems: A brief survey. *IEEE Transactions on Industrial Electronics*, 64(6): 4972–4983, 2017.
- [4] S. Knorn and Z. Chen, Overview: Collective control of multiagent systems. *IEEE Transactions on Control Network Systems*, 3(4): 334–347, 2016.
- [5] W. Zhang, Y. Peng, W. Wei, et al. Real-time conflict-free task assignment and path planning of multi-AGV system in intelligent warehousing. *2018 37th Chinese Control Conference (C-CC)*, Wuhan, China: IEEE, 2018, 5311–5316.
- [6] W. Xie, D. Cabecinhas, R. Cunha, et al. Cooperative path following control of multiple quadcopters with unknown external disturbances. *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, 51(1): 667–679, 2022.
- [7] Z. Peng, J. Wang, D. Wang, et al. An overview of recent advances in coordination control of multiple autonomous surface vehicles. *IEEE Transactions on Industrial Informatics*, 17(2): 732–745, 2021.
- [8] W. Zhang, X. Liu, and P. Han. Progress and challenges of over-water unmanned systems. *Acta Automatica Sinica*, 46(5): 847–857, 2020.
- [9] W. Zhang, N. Wang, S. Wei, et al. Overview of unmanned underwater vehicle swarm development status and key technologies. *Journal of Harbin Engineering University*, 41(2): 289–297, 2019.
- [10] J. Xiang, W. Wei, and Y. Li, Synchronized output regulation of linear networked systems. *IEEE Transactions on Automatic Control*, 54(6): 1336–1341, 2009.
- [11] H. Meng, Z. Chen, L. Zhu, Consensus of a class of second-order nonlinear heterogeneous multi-agent systems with uncertainty and communication delay. *International Journal of Robust and Nonlinear Control*, 26(15): 3311–3329, 2016.
- [12] K. Oh, M. Park, and H. Ahn, A survey of multi-agent formation control. *Automatica*, 53: 424–440, 2015.
- [13] Y. Huang, M. Duan, and L. Mo, Multiagent containment control with nonconvex states constraints, nonuniform time delays, and switching directed networks. *IEEE Transactions on Neural Network Learning Systems*, 31(11): 5021–5028, 2019.
- [14] M. Ji, G. Ferrari-Trecate, M. Egerstedt, et al, Containment control in mobile networks. *IEEE Transactions on Automatic Control*, 53(8): 1972–1975, 2008.
- [15] Z. Chen, A cooperative target-fencing protocol of multiple vehicles. *Automatica*, 107: 591–594, 2019.
- [16] R. Zheng, Z. Lin, M. Fu, et al, Distributed control for uniform circumnavigation of ring-coupled unicycles. *Automatica*, 53: 23–29, 2015.
- [17] L. Shi, R. Zheng, M. Liu, et al, Distributed circumnavigation control of autonomous underwater vehicles based on local information. *Systems & Control Letters*, 148: 104873, 2021.

- [18] X. Yu, J. Ma, N. Ding, et al, Cooperative target enclosing control of multiple mobile robots subject to input disturbances. *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, 51(6): 3440–3449, 2021.
- [19] B. Hu, H. Zhang, Bearing-only motional target-surrounding control for multiple unmanned surface vessels. *IEEE Transactions on Industrial Electronics*, 69(4): 3988–3997, 2022.
- [20] Z. Yu, Y. Zhang, B. Jiang, et al, Fault-tolerant time-varying elliptical formation control of multiple fixed-wing UAVs for cooperative forest fire monitoring. *Journal of Intelligent & Robotic Systems*, 101: 48, 2021.
- [21] MA L, Cooperative target tracking in elliptical formation. *2019 24th International Conference on Methods and Models in Automation and Robotics (MMAR)*, Miedzyzdroje, Poland: IEEE, 2019, 8: 58–63.
- [22] S. Chun and Y. Tian, Multi-targets localization and elliptical circumnavigation by multi-agents using bearing-only measurements in two-dimensional space. *International Journal of Robust and Nonlinear Control*, 30(8): 3250–3268, 2020.
- [23] S. Cao, R. Li, Y. Shi, et al, Safe convex-circumnavigation of one agent around multiple targets using bearing-only measurements. *Automatica*, 134: 109934, 2021.
- [24] L. Chen, Y. Lv, C. Li, et al, Cooperatively surrounding control for multiple Euler-Lagrange systems subjected to uncertain dynamics and input constraints. *Chinese Physics B*, 25(12): 128701, 2016.
- [25] B. Hu, Z. Chen, H. Zhang, Distributed moving target fencing in a regular polygon formation. *IEEE Transactions on Control of Network Systems*, 9(1): 210–218, 2022.
- [26] L. Kou, Z. Chen, J. Xiang, Cooperative fencing control of multiple vehicles for a moving target with an unknown velocity. *IEEE Transactions on Automatic Control*, 67(2): 1008–1015, 2022.
- [27] L. Kou, Y. Huang, Z. Chen, et al, Cooperative fencing control of multiple second-order vehicles for a moving target with and without velocity measurements. *International Journal of Robust and Nonlinear Control*, 31(10): 4602–4615, 2021.
- [28] B. Liu, Z. Chen, H. Zhang, et al, Collective dynamics and control for multiple unmanned surface vessels. *IEEE Transactions on Control Systems Technology*, 28(6): 2540–2547, 2020.
- [29] Y. Shtessel, I. Shkolnikov, and A. Levant, Smooth second-order sliding modes: Missile guidance application. *Automatica*, 43: 1470–1475, 2007.
- [30] S. He, L. Kou, Y. Li and J. Xiang, Robust orientation-sensitive trajectory tracking of underactuated autonomous underwater vehicles. *IEEE Transactions on Industrial Electronics*, 68(9): 8464–8473, 2021.
- [31] S. He, L. Kou, Y. Li and J. Xiang, Position tracking control of fully-actuated underwater vehicles with constrained attitude and velocities. *IEEE Transactions on Industrial Electronics*, 68(12): 13192–13202, 2022.